Q1. JEE Main 2026 (28 January Shift 2)

Given below are two statements :

Statement I : The increasing order of boiling point of hydrogen halides is $HCl < HBr < HI < HF$.

Statement II : The increasing order of melting point of hydrogen halides is $HCl < HBr < HF < HI$.

In the light of the above statements, choose the correct answer from the options given below :

(1) Statement I is true but Statement II is false

(2) Both Statement I and Statement II are true

(3) Both Statement I and Statement II are false

(4) Statement I is false but Statement II is true

Q2. JEE Main 2026 (22 January Shift 1)

A ' p '-block element (E) and hydrogen form a binary cation $(EH_x)^+$, while EH_3 on treatment with K_2HgI_4 in alkaline medium gives a precipitate of basic mercury(II)amido-iodine. Given below are first ionisation enthalpy values $(kJmol^{-1})$ for first element each from group 13, 14, 15 and 16. Identify the correct first ionisation enthalpy value for element E.

(1) 1312 (2) 1402 (3) 1086 (4) 801

Q3. JEE Main 2026 (28 January Shift 1)

Regarding the hydrides of group 15 elements $EH_3(E = N, P, As, Sb)$, select the correct statement from the following:

A. The stability of hydrides decreases down the group.

B. The basicity of hydrides decreases down the group.

C. The reducing character increases down the group.

D. The boiling point increases down the group.

Choose the correct answer from the options given below:

(1) B & C only (2) A, B, C & D

(3) A & D only (4) A, B & C only

ANSWERS AND SOLUTIONS

1. (2) **2.** (2) **3.** (4)

1. **(2)** Correct order of:

(i) Boiling point: HF > HI > HBr > HCl

So increasing order: HCl < HBr < HI < HF (Statement I is true)

(ii) Melting point: HI > HF > HBr > HCl

So increasing order: HCl < HBr < HF < HI (Statement II is true)

Both Statement I and Statement II are true.

2. **(2)** The p-block element E forms a binary cation $(EH_x)^+$ with hydrogen and produces a basic mercury(II) amido-iodide precipitate with K_2HgI_4 in alkaline medium.

This behavior is characteristic of ammonia (NH_3), which forms the ammonium ion $(NH_4)^+$ and reacts with Nessler's reagent (K_2HgI_4) to form a characteristic brown/yellow precipitate.

Therefore, E is Nitrogen (N), a Group 15 p-block element with electronic configuration $1s^2 2s^2 2p^3$.

Nitrogen has a notably high first ionization enthalpy compared to other group 15 elements due to:

Small atomic radius

High effective nuclear charge

Stable half-filled p^3 subshell configuration

The first ionization enthalpy of nitrogen is 1402 kJ/mol.

3. **(4)** For group 15 hydrides EH_3 (N, P, As, Sb):

Statement A is correct—stability decreases down the group as H-E bond strength decreases.

Statement B is correct—basicity decreases as the central atom becomes less electronegative, making it less willing to share electron pairs.

Statement C is correct—reducing character increases down the group as the hydrides become easier to oxidize.

Statement D is incorrect—boiling points do not consistently increase; NH_3 is anomalously high (-33°C) due to strong hydrogen bonding, but PH_3 (-87°C) is lower, and the trend is irregular.

Therefore, A, B & C only are correct.